Taj Dyson

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Experimental physicist studying axion dark matter using cryogenic quantum devices.

Education

Stanford University Sep. 2021 – Jun. 2027
PhD, Physics (GPA 3.90)
Thesis: *High-volume Resonators for Haloscopes Probing the Post-inflationary Axion*

McGill University Aug. 2018 – May 2021
Bachelor of Science, Honours Physics (GPA 3.76)
Thesis: *Radio Interferometer Synchronization Using External Signal*

Dawson College Aug. 2016 – May 2018
DCS, First Choice Sciences, Honours List

Selected Awards

NSERC Postgraduate Scholarships – Doctoral 2023 – 2026
A federal award granted to approx. 400 high-caliber Canadian students per year.

Robert H. Siemann Fellowship 2022 – 2025
Awarded to around two outstanding physics graduate students working at SLAC.

NSERC Undergraduate Summer Research Award & FRQNT Supplement 2020
A federal award supporting more than 3,000 undergraduates showing academic excellence and research potential.

BLUE Fellowship at McGill's Building 21 2020
Supported my self-directed research on emergent complexity at Building 21 in a cohort of about 20.

McGill Physics Hackathon Winner – Arts & Science 2019
One of eight prizes awarded in this weekend-long event.

Selected Publications

(Click entry to view)

- H. Magoon, T. Aralis, **T. Dyson**, *et. al.* "A First Demonstration of the SQUAT Detector Architecture: Direct Measurement of Resonator-Free Charge-Sensitive Transmons," *In submission for Physical Review Applied* (2026).
- G. Bratrud, S. Lewis, K. Anyang, **T. Dyson**, *et. al.* "Measurement of Correlated Charge Noise in Superconducting Qubits at an Underground Facility," *Nature Communications* (2025)
- **T. Dyson**, C. Bartram, A. Davidson, *et al.*, "High-volume Tunable Resonator for Axion Searches Above 7 GHz," *Phys. Rev. Applied* (2024).
- **T. Dyson**, H. C. Chiang, E. Egan, *et al.*, "Radio-Frequency Interference at the McGill Arctic Research Station," *Journal of Astronomical Instrumentation* (2021).

Selected Presentations and Outreach

(Click entry to view, see below for full list)

Design and Characterization of SQUAT Detectors Mar. 2026
Talk, APS Global Physics Summit

High-frequency Haloscopes for the ADMX-VERA Program Mar. 2025
Talk, APS Global Physics Summit

Resonating with Dark Matter with ADMX-VERA Dec. 2024
Article, Stanford KIPAC Research Highlights

Research Experience

Graduate Research Assistant – Axion Haloscope Sep. 2021 – Present

Stanford University, supervised by Prof. Chao-Lin Kuo

- Characterized the resonances of a novel prototype haloscope resonator for axion and dark photon dark matter detection.
- Developed an algorithm to automatically align the haloscope across five degrees of freedom based on measurements possible at cryogenic temperatures.
- Performed precision metrology on haloscope parts in a custom-built LN_2 bath.
- Designed and commissioned a cryogenic dark photon search featuring a 7-kg moving part with six degrees of freedom, including a vacuum system extension, thermal sinking, and radiation shielding.
- Performed a tabletop dark photon search using custom control software and a SQL database, with analysis using Python. The forthcoming results will set leading constraints on dark photons with mass 21–29 μeV.
- Mentored four undergraduates, with weekly meetings to set goals and assess challenges as well as collaborative work in the laboratory.

Graduate Research Assistant – Qubit-Based Sensors Mar. 2022 – Present

SLAC and Stanford University, supervised by Dr. Noah Kurinsky

- Took continuous-wave and pulsed measurements to characterize superconducting quasiparticle-amplifying transmons (SQUATs), a novel low-energy rare event detector.
- Simulated SQUAT data to optimize the choice of integration time for given device properties.
- Designed and integrated a hardware system for actuating cryogenic switches for radio frequency cables, and wrote a Python library to interface with it. These switches enable new types of cryogenic measurement and allow multiple devices to be tested without cycling the cryostat.
- Characterized a superconducting traveling wave parametric amplifier (TWPA), finding optimal operating parameters and its noise temperature.
- Learned firsthand to operate a dilution refrigerator.

Graduate Research Assistant – Axion Dark Matter Experiment Feb. 2023 – Present

Stanford University, supervised by Prof. Chao-Lin Kuo for the ADMX Collaboration

- Sped up analysis of data taken in high-resolution mode by employing pre-compiled size- and type-aware functions rather than interpreted Python.
- Coordinated across the collaboration to gather past simulations and measurements done for the proposed ADMX Extended Frequency Range experiment, and led the writing of the design study.
- Operated the ADMX main experiment during science data taking, including Josephson parametric amplifier pump optimization and rod actuation for re-scanning.

Graduate Research Assistant – Cosmic Microwave Background Receivers May 2022 – Jul. 2023

Stanford University, supervised by Prof. Chao-Lin Kuo

- Took cryogenic measurements of the noise performance of a novel detector type, thermal kinetic inductance detectors (TKIDs).
- Contributed to design the cryogenic radio-frequency readout chain for a receiver in the BICEP array.

Undergraduate Research Assistant – Radio Cosmology Field Work Sep. 2018 – Jul. 2021

McGill University, supervised by Prof. Cynthia Chiang

- Developed and deployed solar and wind power solutions for radio interferometer stations in remote locations such as Uapishka Station and the McGill Arctic Research Station.
- Designed & built electronic devices and wrote C++ Arduino firmware for power control & logging.
- Flagged radio-frequency interference in radio data using custom Python scripts.

Complete Presentations

Conference Talks

Design and Characterization of SQUAT Detectors Mar. 2026

APS Global Physics Summit

High-frequency Haloscopes for the ADMX-VERA Program Mar. 2025

APS Global Physics Summit

Demonstration of a High-Volume, Tunable Haloscope Above 7 GHz Jul. 2023

Patras Workshop on Axions, WIMPs, and WISPs

Seminars

Cryogenic Characterization of a Tunable, High-Volume Haloscope — Dec. 2026
LBNL Institute for Nuclear and Particle Astrophysics Seminar (invited) — *(upcoming)*

Axion Dark Matter eXperiment – Volume-Enhanced Resonators for Axions (ADMX-VERA) — Jan. 2026
Berkeley Axion Works Biweekly Seminar (invited)

Developing ADMX-VERA, an Axion Haloscope Above 7 GHz — Mar. 2024
Fermilab Superconducting Quantum Materials and Systems Seminar (invited)

Collaboration Talks

First-principles Downtime Estimation for ADMX Extended Frequency Range — Apr. 2024
ADMX Collaboration Meeting

ADMX-VERA R&D Update — Apr. 2024
ADMX Collaboration Meeting

Axion Activities at Stanford — Feb. 2023
ADMX Collaboration Meeting

Radio Frequency Cable Routing in BICEP 4 — Oct. 2021
BICEP Collaboration Meeting

Local Talks

Axion Search Experiments — Jul. 2025
Dark Matter/Quantum Information Science Meeting

Does Dark Matter Have to Exist? — Mar. 2023
Dark Matter/Quantum Information Science Meeting

Traveling Wave Parametric Amplifiers with Josephson Junctions and Resonator Phase-matching — May 2022
Quantum Sensing Journal Club

The Axion Project: Story So Far — Nov. 2021
Meeting of Astrophysics Students at Stanford

Posters

Dark Photon Searches with ADMX-VERA Single-Wedge Resonators — May 2025
Berkeley Axion Dark Matter Workshop

An Overview of the Axion Dark Matter eXperiment's Volume-Enhanced Resonators for Axions — Mar. 2025
UCLA International Symposium on Sources and Detection of Dark Matter and Dark Energy in the Universe

Outreach

Public Talks

Radio-Frequency Interference at the McGill Arctic Research Station — Sep. 2019
McGill Soup and Science Public Talks

Articles

Resonating with Dark Matter with ADMX-VERA — Dec. 2024
Stanford KIPAC Research Highlights

ALBATROS radio astronomy Product Showcase — Sep. 2019
MagPi Magazine Article (interviewed)

Educational Programs

Volunteer Course Designer and Instructor — Jun. 2023 – Jun. 2024
Virtù Institute at Stanford University

Teaching Experience

Teaching Assistant — Sep. 2023 – Dec. 2023
Stanford University — *Introduction to Heat & Light*

Teaching Assistant — Jan. 2023 – Mar. 2023
Stanford University — *Introduction to Electricity and Magnetism*

Teaching Assistant — Jan. 2022 – Mar. 2022
Stanford University — *Introduction to Mechanics*

TEAM Undergraduate TA — Jan. 2021 – May 2021
McGill University — *Advanced Undergraduate – Data Science and Observational Astrophysics*

Instructor — Feb. 2017 – Sep. 2017
Kids Code Jeunesse — *Python, HTML, Scratch*

Academic Service

Camping Czar — Jun. 2022 – May 2023
Stanford University Graduate Students of Applied Physics and Physics

Vice President — Sep. 2023 – Nov. 2024
Stanford University Graduate Students of Applied Physics and Physics

Technical Proficiencies

Laboratory: Dilution refrigerator operation, vector network analyzer (VNA) measurement, spectrum analyzer measurement, electronics chain design and assembly, high-vacuum system assembly and helium leak checking
Manufacturing: Manual mill, manual lathe, waterjet cutter, laser cutter/engraver, soldering iron, Dremel rotary tool, finishing tools, 3D printer, coordinate measuring machine (CMM) for metrology, wet bench for copper passivation
Design & Simulation: SOLIDWORKS, KiCAD, COMSOL, HFSS
Software tools: Claude Code, SQL databases, Pyvisa experiment control, local networking
Languages: Python, bash, Perl, C, C++, Lua, Julia

Complete Publications

T. Dyson, *et al.*, "A Polarization-optimized Search for Dark Photons from 29.4 to 33.1 μeV" *In preparation.*

ADMX Collaboration (incl. **T. Dyson**), "Axion Dark Matter eXperiment with Extended Frequency Range (ADMX-EFR) – Conceptual Design and Sensitivity" *In preparation.*

H. Magoon, *et al.* (incl. **T. Dyson**), "A First Demonstration of the SQUAT Detector Architecture: Direct Measurement of Resonator-Free Charge-Sensitive Transmons," *arXiv:2601.16261* (2026).

H. Magoon, *et al.* (incl. **T. Dyson**), "A First Demonstration of the SQUAT Detector Architecture: Direct Measurement of Resonator-Free Charge-Sensitive Transmons," *arXiv:2601.16261* (2026).

C. Goodman, *et al.* (incl. **T. Dyson**), "ADMX Axion Dark Matter Bounds around 3.3 μeV with Dine-Fischler-Srednicki-Zhitnitsky Discovery Ability," *Phys. Rev. Lett.* (2025).

ADMX Collaboration (incl. **T. Dyson**), "Search for Axion Dark Matter from 1.1 to 1.3 GHz with ADMX," *Phys. Rev. Lett.* (2025).

G. Bratrud, *et al.* (incl. **T. Dyson**), "Measurement of Correlated Charge Noise in Superconducting Qubits at an Underground Facility," *Nature Communications* (2025).

A. T. Hipp, *et al.* (incl. **T. Dyson**), "Search for Non-virialized Axions with 3.3–4.2 μeV Mass at Selected Resolving Powers," *Phys. Rev. D* (2025).

C. L. Kuo, *et al.* (incl. **T. Dyson**), "Maximizing Quantum Enhancement in Axion Dark Matter Experiments," *Phys. Rev. D* (2025).

M. Guzzetti, *et al.* (incl. **T. Dyson**), "Improved Receiver Noise Calibration for ADMX Axion Search: 4.54 to 5.41 μeV," *Phys. Rev. D* (2025).

T. Dyson, *et al.*, "High-volume Tunable Resonator for Axion Searches Above 7 GHz," *Phys. Rev. Applied* (2024).

T. Dyson, *et al.*, "Radio-Frequency Interference at the McGill Arctic Research Station," *Journal of Astronomical Instrumentation* (2021).